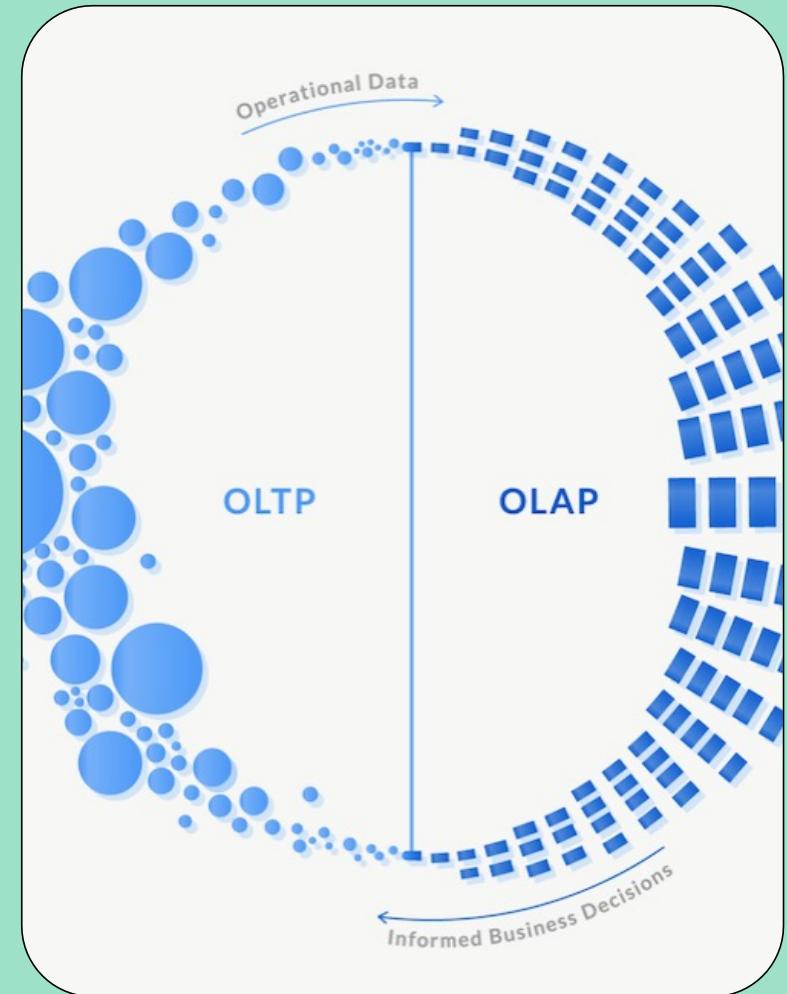


Project 3 & 4

Administration
OLTP to OLAP



Source Dataset



- Education Data
- Format: CSV
- No. of files: 19
- Total size: 1.3 GB

```
applicant_id  
applicant_name  
applicant_gender  
applicant_dob  
applicant_country  
applicant_qual_test_score  
applicant_high_school_GPA  
scholarship_grade  
scholarship_pct  
interview_date  
interview_score  
admission_date  
institute_id  
course_name
```

Applicant Data

```
institute_id  
institute_name institute_fee  
institute_reputation  
institute_campus_job_placement_pct  
institute_campus_area  
score_cut_off  
total_no_of_students  
applied_no_of_students  
declined_no_of_student_pct  
accepted_no_of_student_pct
```

Institute Data

Administration



- Create 2 tables – Applicant and Institute, with the following properties
 - Applicant Table – DistKey: applicant_id, SortKey: applicant_id, DistStyle: EVEN, PrimaryKey: applicant_id, ForeignKey: institute_id
 - Institute Table - DistKey: institute_id, SortKey: institute_id, DistStyle: KEY, PrimaryKey: institute_id
- Load each applicant file 4 times into the Applicant table. Put all files in one and then load again. Is there any difference in execution time.
- Check the following:
 - What is the size of individual tables.
 - What is the slice distribution per node. How many blocks are there per slice.
 - Find DISTKEY, SORTKEY and DISTSTYLE of each table.
 - Find if column encodings have been applied in the tables.

Administration - observation

- You realize after a few weeks that most of the queries have the following patterns (priority is given along with the statements):
 - Find all applicants that have been placed in individual institutes. P1
 - Find information about individual applicants/students. P1
 - No. of students per institute. P2
 - Details of an institute, which are queried by students/applicants. P2
 - All students that have been declined from an institute. P3
 - Age, country and qualification score of individual applicants. P4
 - Reputation of an institute along with its score cut off. P5
 - All applicants with interview score more than certain percentage. P5
 - Which institutes have the highest decline percentage. P5

Administration



- Based on the query patterns in the previous slide, please do the following, if applicable:
 - Should DistKey and SortKey should be same for both the tables. If not then modify them.
 - What should be the primary key. Modify if necessary.
 - Can interleaved sort key be leveraged in any of the cases.
- Please check if there are duplicate rows. If yes, delete or remove all duplicate rows in applicant table.
- Check the size of the table before and after removing duplicate rows. Fix it if necessary.
- Pls check how many times the tables were loaded. Were there any errors.

Administration



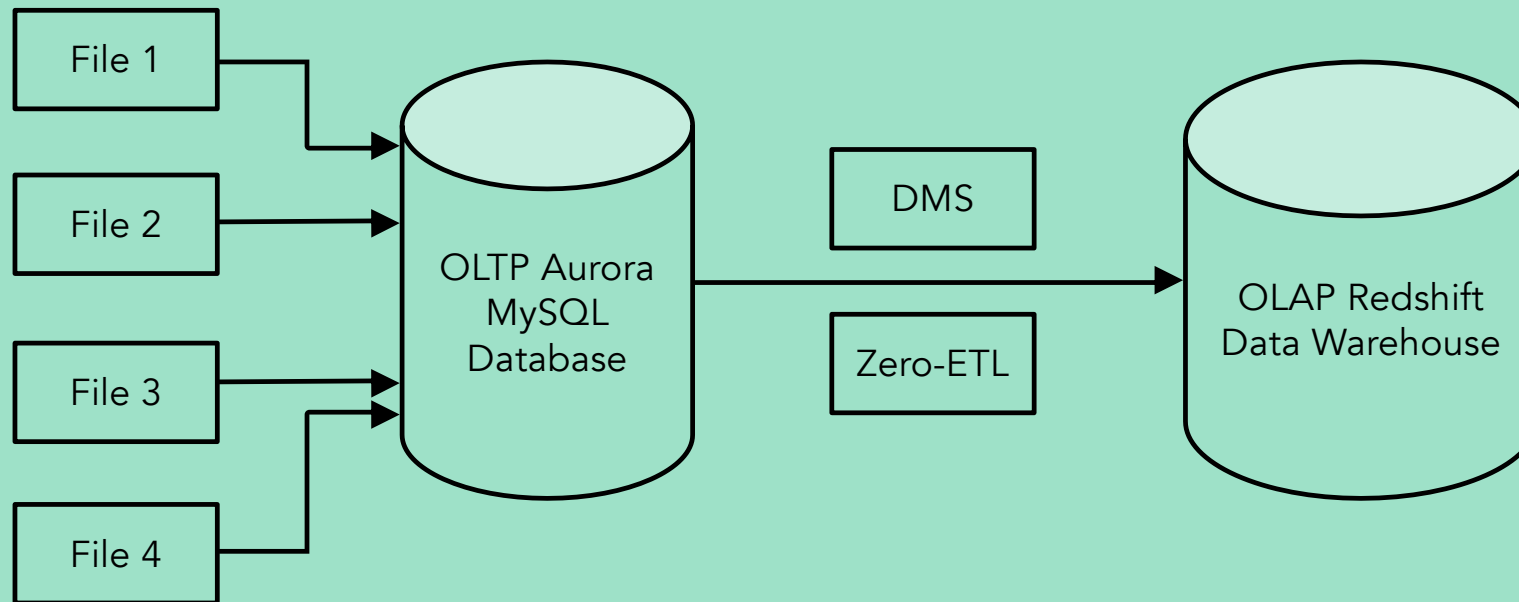
- Create Orders table and load using large orders files. Check the status of load every 2 mins.
- Split orders files into a total of 20 files and put them in separate S3 locations. Load them using manifest files.
- Unload orders that are older than 2 years and delete them from the table. Keep the older records in S3.

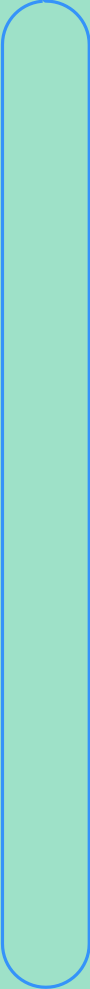
OLTP to OLAP



- There's an OLTP database in Aurora MySQL, where applicant data is loaded on a regular basis.
- Load the applicant files one-by-one every 5 mins. But the records should be automatically be loaded to Redshift from Aurora.
- 2 options to load data from OLTP to OLAP. Use any one or both.
 - Zero ETL
 - AWS DMS

OLTP to OLAP





Thank you



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